

TTK4240 Industriell elektroteknikk

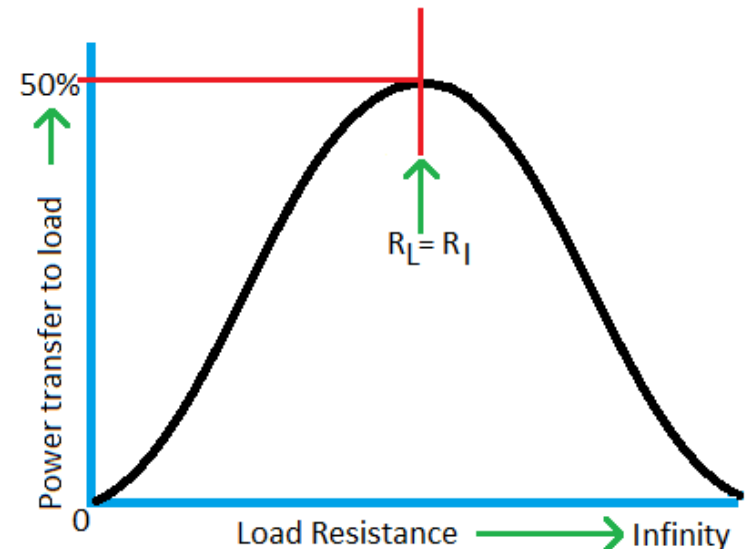
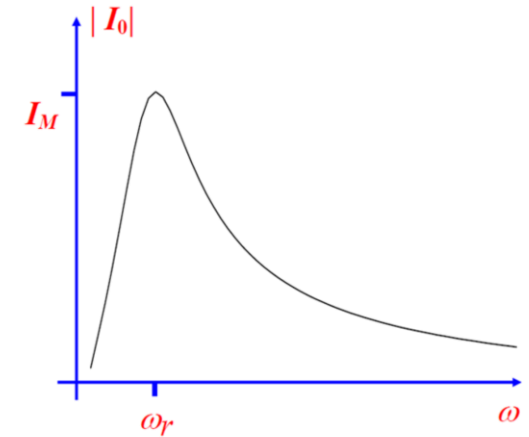
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Marta Molinas

Institutt for teknisk kybernetikk

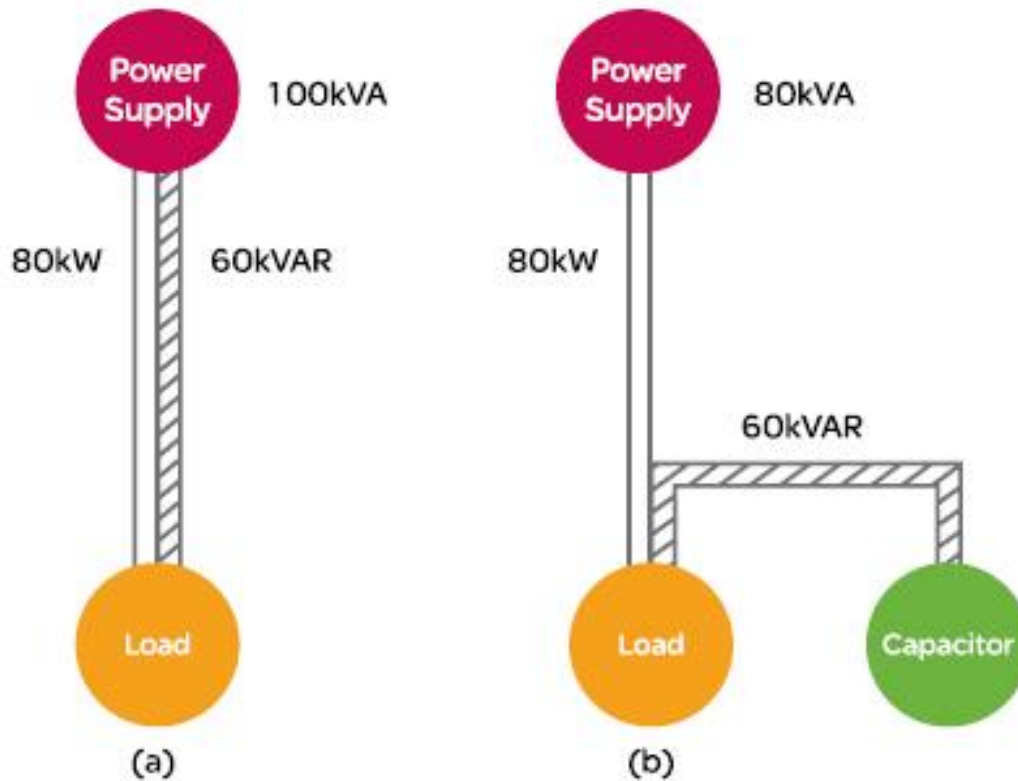
Lecture - Week 05 – Part I

- Current, voltage and power calculations
 - Complex power **[S]**
 - Maximum real power delivered to a load
 - Load impedance required to deliver maximum real power to the load
- Exercises
 - Complex Power
 - Maximum Power Transfer
 - Restrictive loads



Reactive power

Review



Rating of power supply:

- 100 kVA w/o Capacitor
- 80 kVA w/ Capacitor

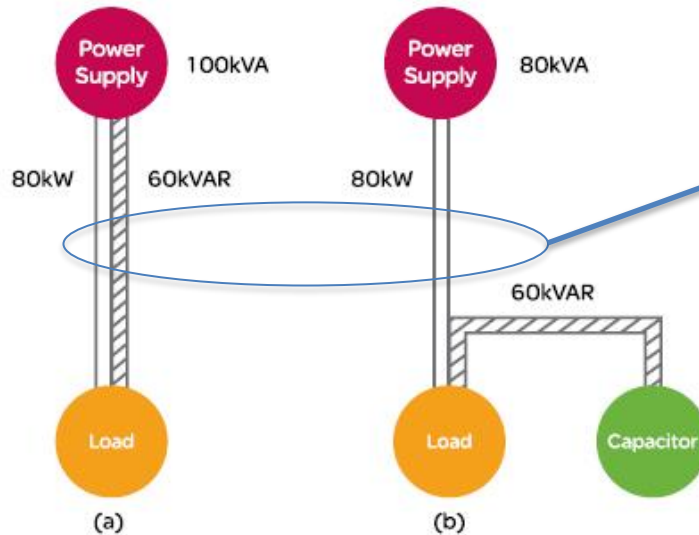
Apparent power of transmission line:

$$S = \sqrt{80^2 + 60^2} = 100kVA \quad S = 80kVA$$

Without compensation,
The rating of transmission
line should be increased.

Reactive Power

Review



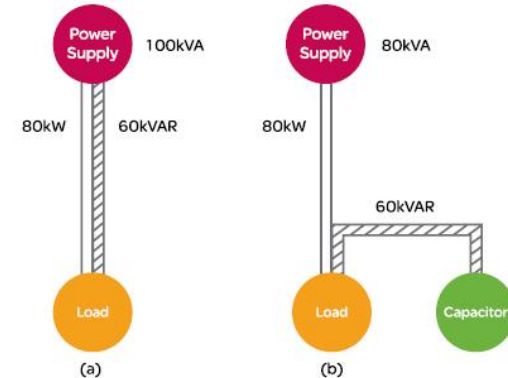
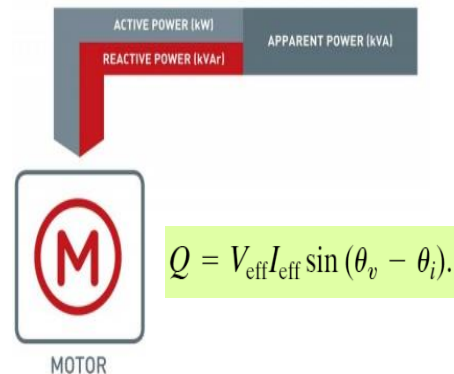
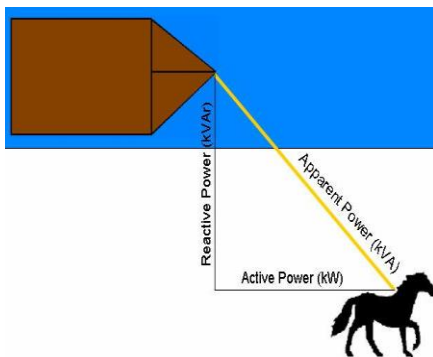
The transmission system usually is not ideal and exists resistance R_{loss} .

$$P_{loss} = |\mathbf{I}|^2 R_{loss}$$

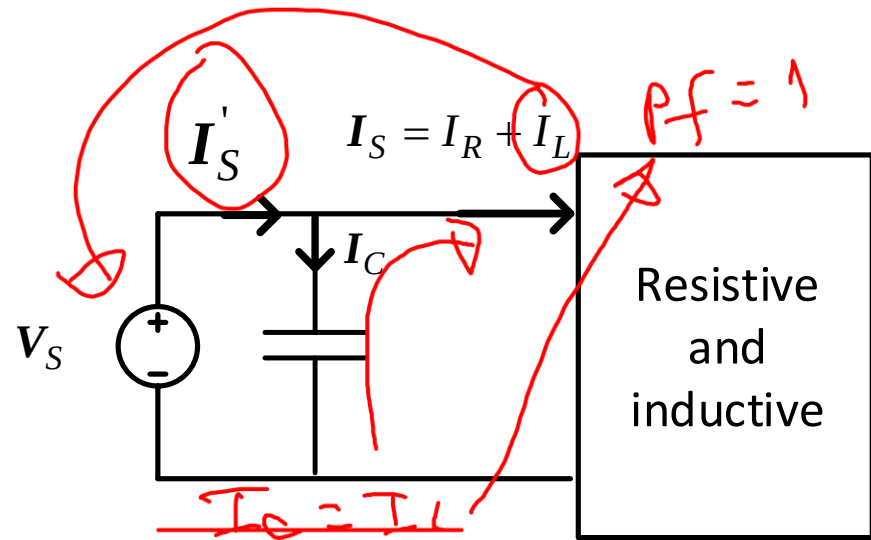
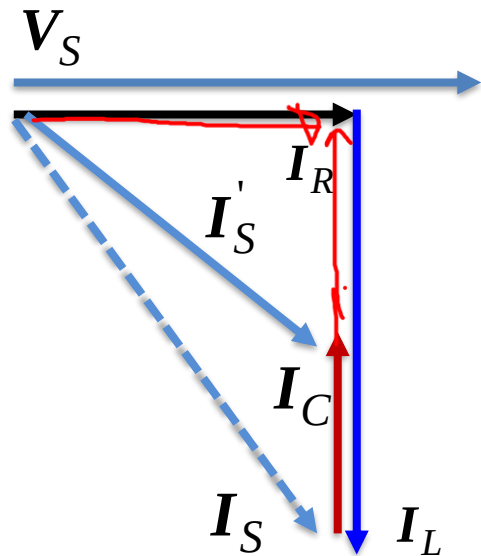
*The reactive power can increase the amount of **apparent power** flowing between the source and load, thus usually leading to more losses.*

Feature Summary of Reactive power

- Reactive power does not **contribute to the actual work** (its average over one period is zero).
- The total **rating of an equipment** should include the reactive power consumption
- Since it flows back and forth between the source and load, if the **transmission is nonideal**, it will lead to more losses.



Example: Reactive power compensation Effect



Question: What is the compensation effect ?

$$|I'_S| < |I_S|$$

Implies a fewer transmission losses

$$P_{loss} = |I|^2 R_{loss}$$

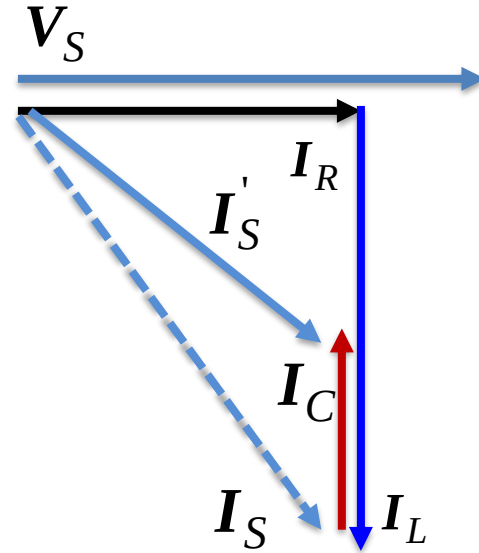
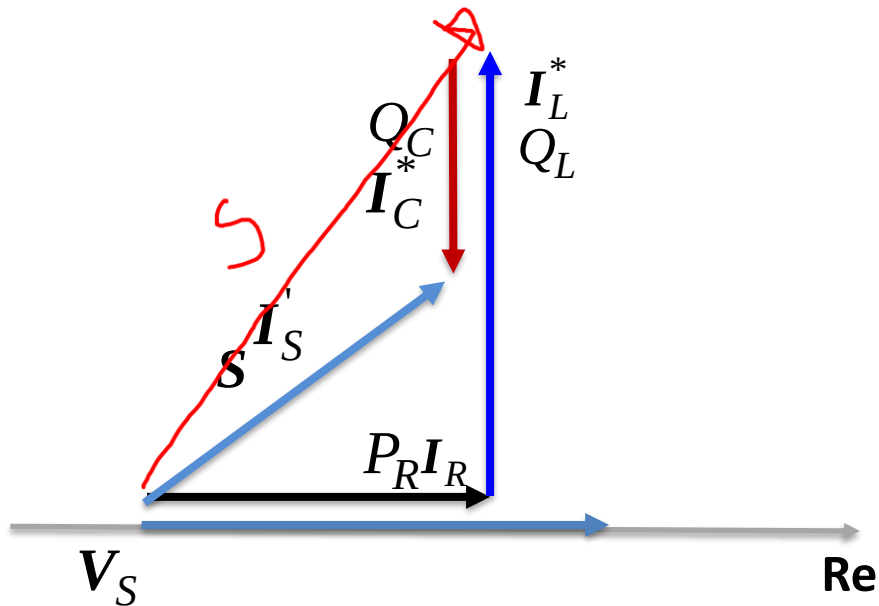
What is **power relationship** after the compensation?

Before

$$\mathbf{S} = \mathbf{V}_S \mathbf{I}_S^*$$

After

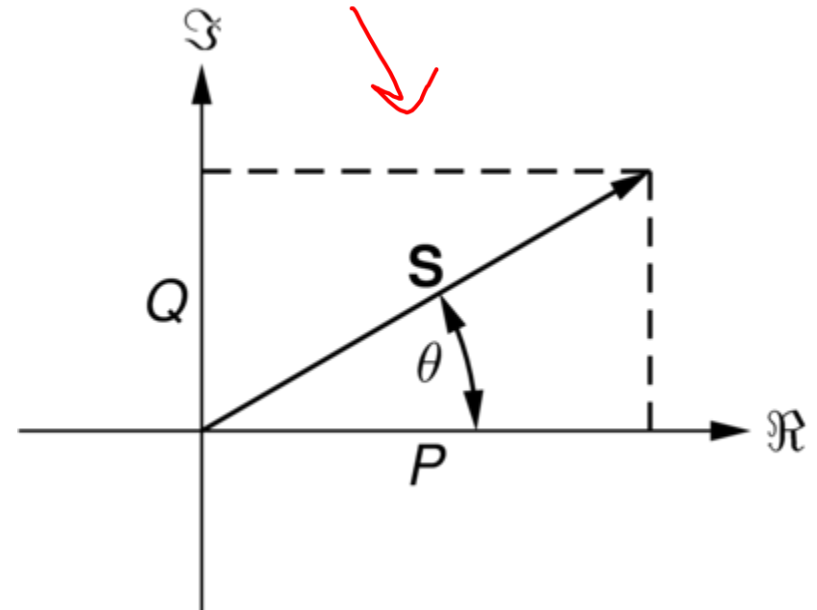
$$\mathbf{S} = \mathbf{V}_S \left(\mathbf{I}_S' \right)^*$$



Complex Power

$$S = P + jQ.$$

$$S = V_{eff} \cdot I_{eff}^*$$



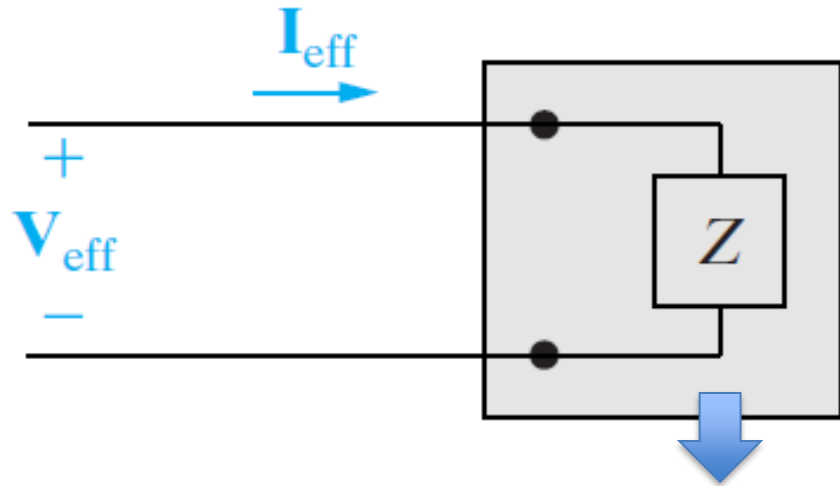
Variants of Complex Power

$$S = V_{eff} \cdot I_{eff}^*$$

Two variants of complex power

$$S = \frac{|V_{eff}|^2}{Z^*}$$

$$S = Z |I_{eff}|^2$$



Impedance

$$Z = \left| \frac{V_{eff}}{I_{eff}} \right| e^{j(\theta_v - \theta_i)}$$

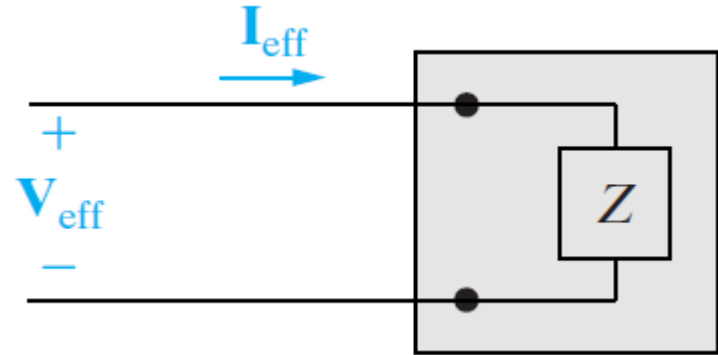
$X > 0, \text{inductive}$
 $X < 0, \text{capacitive}$

$$= \underbrace{\left| \frac{V_{eff}}{I_{eff}} \right| \cos(\theta_v - \theta_i)}_R + j \underbrace{\left| \frac{V_{eff}}{I_{eff}} \right| \sin(\theta_v - \theta_i)}_X$$

Variants of Active and Reactive Power-I

$$Z = R + jX$$

$$S = \mathbf{Z} |I_{eff}|^2 = P + jQ$$



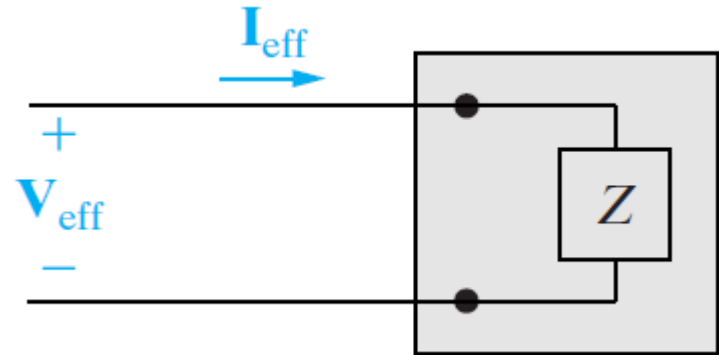
$$\mathbf{P} + j\mathbf{Q} = |I_{eff}|^2 (R + jX)$$

$$P = |I_{eff}|^2 R$$

$$Q = |I_{eff}|^2 X$$

Variants of Active and Reactive power-II

$$S = \frac{|V_{eff}|^2}{\mathbf{Z}^*} = \underline{P + jQ}$$



$$\mathbf{Z}^* = \text{conj}\{R + jX\} = \underline{R - jX}$$

$$P + jQ = \frac{|V_{eff}|^2 \mathbf{Z}}{\mathbf{Z}^* \mathbf{Z}} = \frac{|V_{eff}|^2 (R + jX)}{|\mathbf{Z}|^2}$$

$$P = \frac{|V_{eff}|^2}{|\mathbf{Z}|^2} R$$

$$Q = \frac{|V_{eff}|^2}{|\mathbf{Z}|^2} X$$

Apparent Power

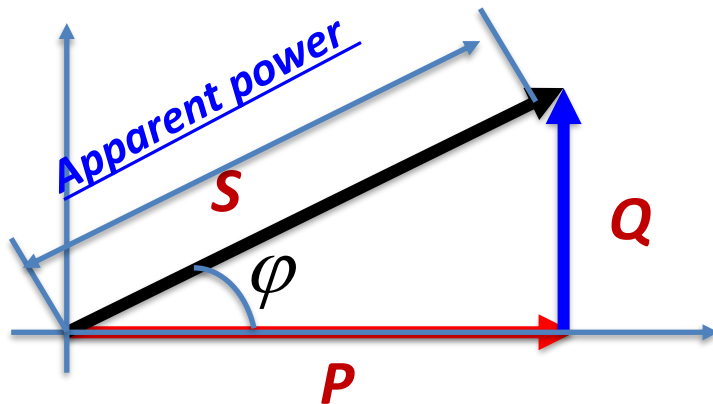
complex power

$$S = \frac{1}{2} \mathbf{V} \mathbf{I}^*$$

peak
phasor
values

Apparent power =

Magnitude of the complex power S



The same derivation technique applied to this expression of S will give an alternative expression based on peak phasor values.

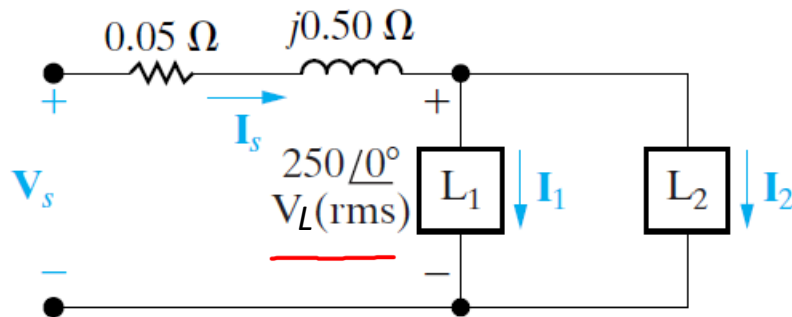
Practical Example

Calculate Power in parallel loads

L1 absorbs average power of 8 kW at leading power factor 0.8 $\rightarrow C$

L2 absorbs 20 kVA at lagging power factor 0.6 $\rightarrow L$

- Calculate the power factor of the two loads in parallel
- What is the required apparent power to supply the loads, the current I_s and the average power loss in the transmission line?
- If frequency is 60 Hz, which value of capacitor in parallel with the loads will correct the power factor to 1?



$$S = V_{eff} I_{eff}^*$$

$$X = \frac{|V_{eff}|^2}{Q}$$

$$C = \frac{-1}{\omega X}$$

$$|S| = P$$

$Pf = 1$

Answer:

$pf = \cos(0 + 26.57^\circ) = 0.8944$ lagging.

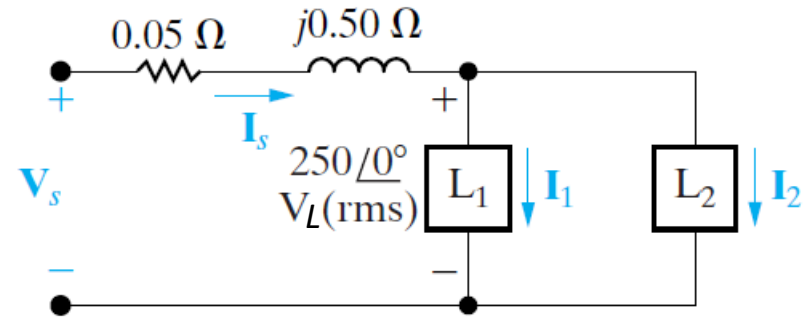
$|S| = |20 + j10| = 22.36$ kVA.

$424.4 \mu F$.

L1 absorbs average power of 8 kW at leading power factor 0.8
 L2 absorbs 20 kVA at lagging power factor 0.6

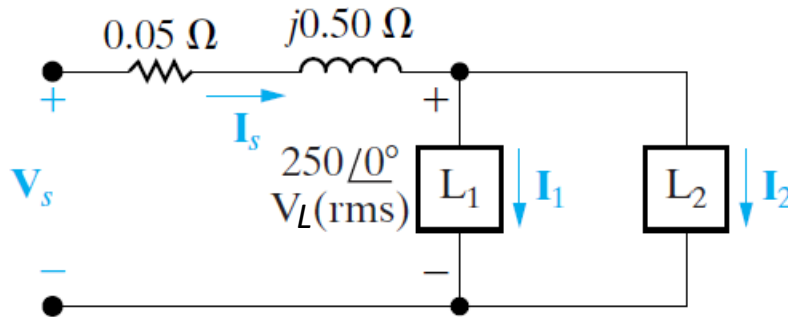
L1: $P_{L1} = +8\text{kW}$ *Leading*
 $\cos \varphi_{L1} = 0.8$ $\varphi_{L1} < 0$

L2: $S_{L2} = 20\text{kVA}$ *Lagging*
 $\cos \varphi_{L2} = 0.6$ $\varphi_{L2} > 0$



- Calculate the power factor of the two loads in parallel?
- What is the required apparent power to supply the loads, the current I_s and the average power loss in the transmission line?
- If frequency is 60 Hz, which value of capacitor in parallel with the loads will correct the power factor to 1?

a) Calculate the power factor of the two loads in parallel ?



L1: $P_{L1} = +8kW$ $\cos \varphi_{L1} = 0.8$

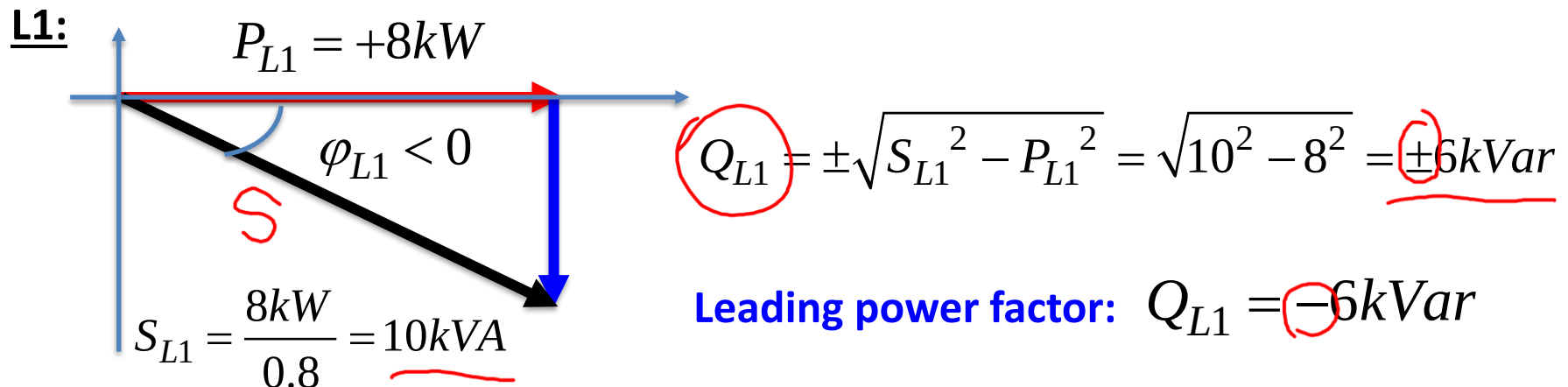
$\varphi_{L1} < 0$

L2: $S_{L2} = 20kVA$ $\cos \varphi_{L2} = 0.6$

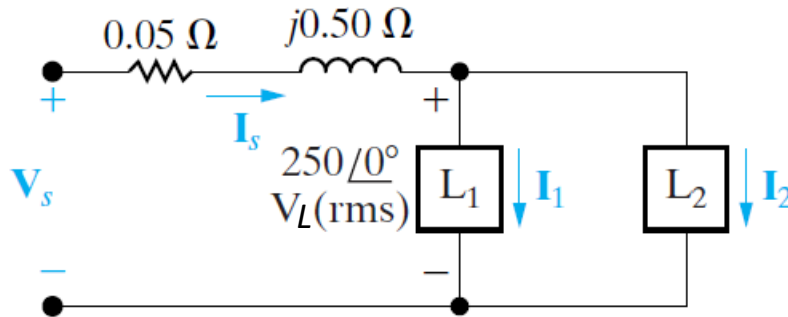
$\varphi_{L2} > 0$

Method#1: using the voltage and current phasor

Method#2: using the power relationship



a) Calculate the power factor of the two loads in parallel ?



L1: $P_{L1} = +8kW$ $\cos \varphi_{L1} = 0.8$

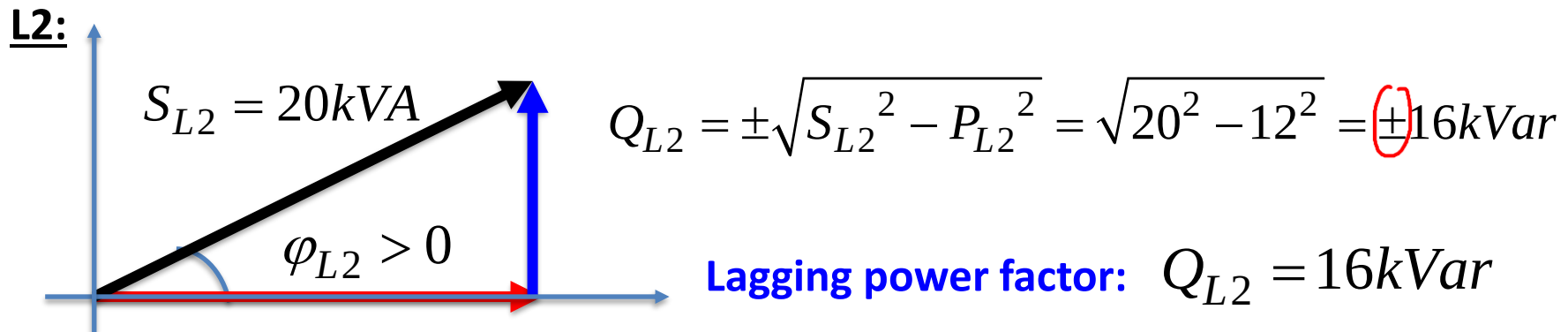
$\varphi_{L1} < 0$

L2: $S_{L2} = 20kVA$ $\cos \varphi_{L2} = 0.6$

$\varphi_{L2} > 0$

Method#1: using the voltage and current phasor

Method#2: using the power relationship



$$P_{L2} = S_{L2} \cos \varphi_{L2} = 20 \cdot 0.6 = 12kW$$

a) Calculate the power factor of the two loads in parallel ?

The complex power L1

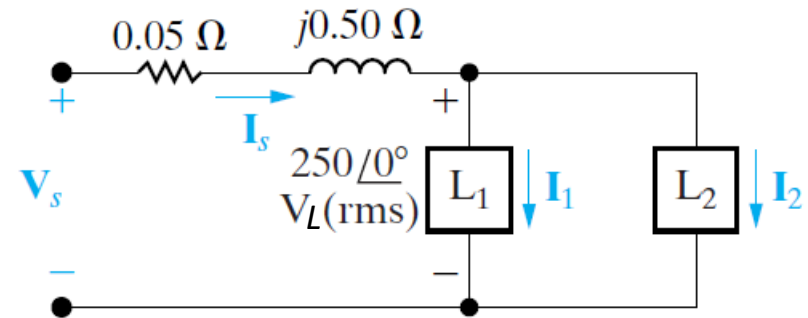
$$\begin{aligned} S_{L1} &= P_{L1} + jQ_{L1} \\ &= \underline{(8 - 6j) \text{ kVA}} \end{aligned}$$

The complex power L2

$$\begin{aligned} S_{L2} &= P_{L2} + jQ_{L2} \\ &= \underline{(12 + 16j) \text{ kVA}} \end{aligned}$$

$$\varphi_L = \tan^{-1} \frac{10}{20} = 26.56^\circ > 0$$

Lagging



The complex power L1+L2

$$\begin{aligned} S_L &= S_{L1} + S_{L2} \\ &= \underline{(20 + 10j) \text{ kVA}} \end{aligned}$$

$$\underline{\cos \varphi_L} = \cos 26.56^\circ = \underline{0.894}$$

b) What is the required **apparent power** to supply the loads, the **current I_s** and the **average power loss** in the transmission line?

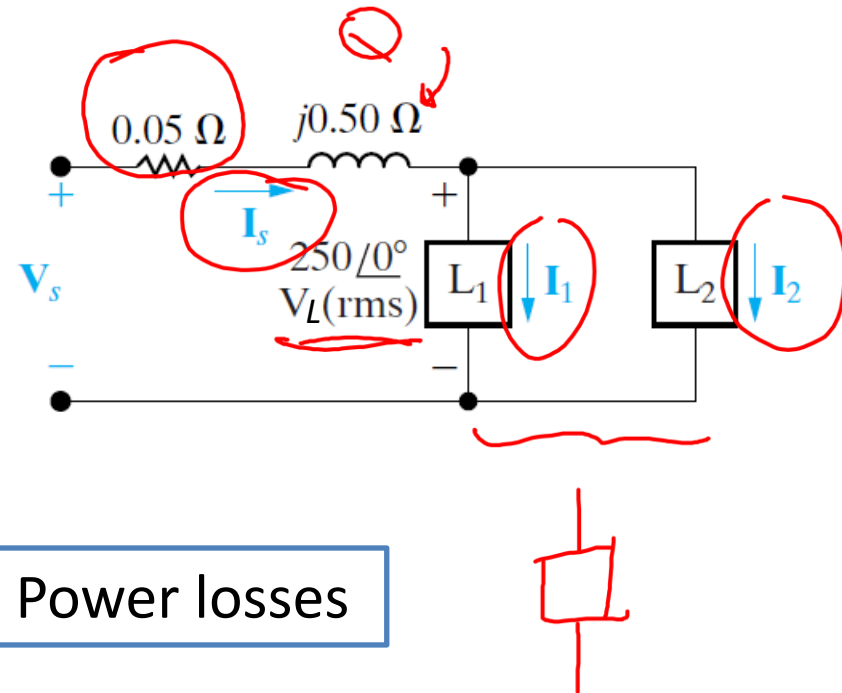
The complex power L1+L2

$$\mathbf{S}_L = (20 + 10j) \text{ kVA}$$

$$|\mathbf{S}_L| = \sqrt{20^2 + 10^2} = \underline{\underline{22.36 \text{ kVA}}}$$

The load current phasor

$$\begin{aligned} \mathbf{I}_L = \mathbf{I}_1 + \mathbf{I}_2 = \mathbf{I}_S &= \left(\frac{\mathbf{S}_L}{\mathbf{V}_L} \right)^* \\ &= \frac{(20 - 10j) \cdot 10^3}{250} = \underline{\underline{89.44 \angle -26.56^\circ}} \end{aligned}$$



Power losses

$$\begin{aligned} P_{loss} &= \underline{R_{loss}} |\mathbf{I}_S|^2 \\ &= \underline{0.05} \cdot \underline{|89.44|^2} = \underline{\underline{0.4 \text{ kW}}} \end{aligned}$$

c) If frequency is 60 Hz, which value of capacitor in parallel with the loads will correct the power factor to 1?

$$\cos \varphi_L = 0.894$$

$$Q_L + Q_C = 0$$

Before compensation

$$S_L = P_L + jQ_L = (20 + 10j) \text{ kVA}$$

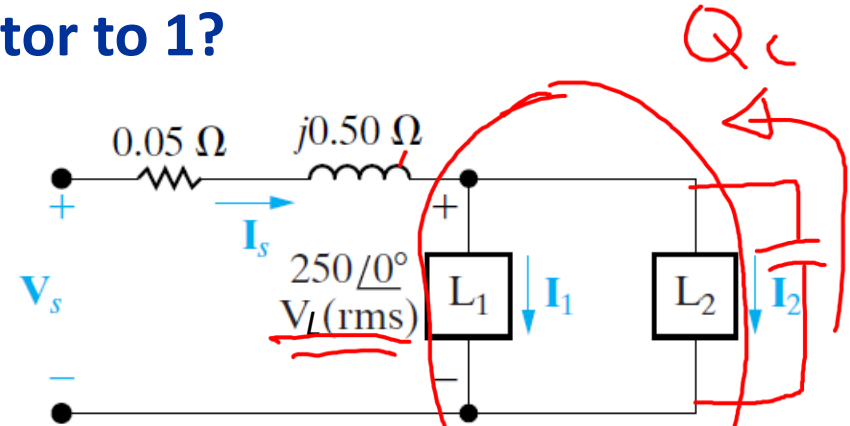
$$Q_C = -Q_L = -10 \text{ kVar}$$

$$S = \frac{|V_{eff}|^2}{Z^*} \quad Z = -j \frac{1}{\omega C}$$

$$S = j \left(-\omega C |V_{eff}|^2 \right)$$

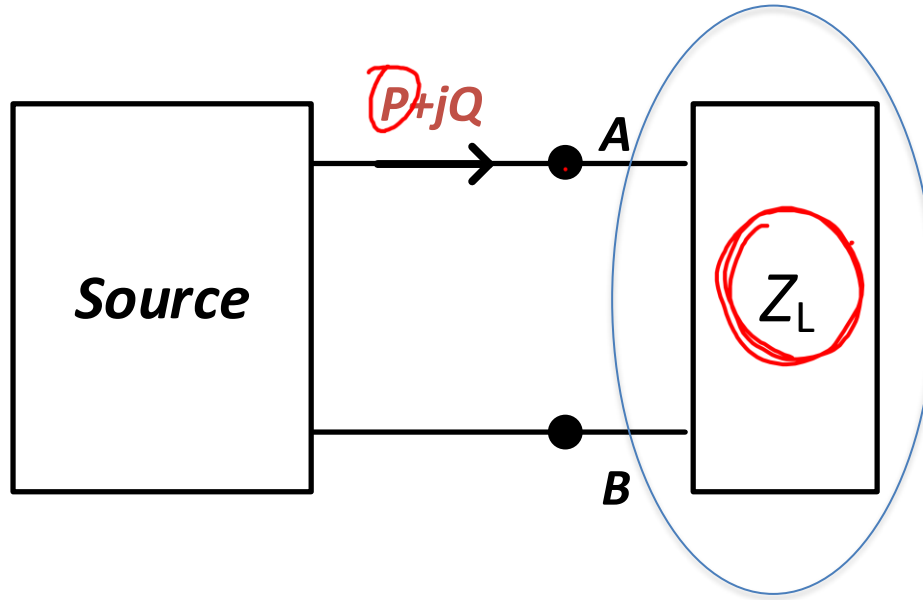
$$-\omega C |V_{eff}|^2 = -10 \text{ kVar}$$

$$C = \frac{10 \text{ kVar}}{\omega |V_{eff}|^2} = \frac{10 \cdot 10^3}{60 \cdot 2 \cdot \pi |250|^2} = 424.6 \mu\text{F}$$



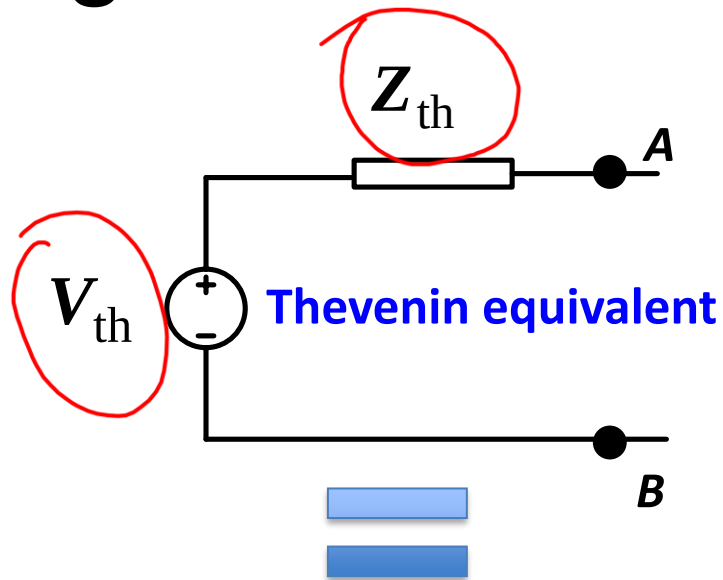
Maximum Power Transfer

Maximum Power Transfer



Question: What is the load impedance Z_L that results in the delivery of maximum average power to terminals a and b.

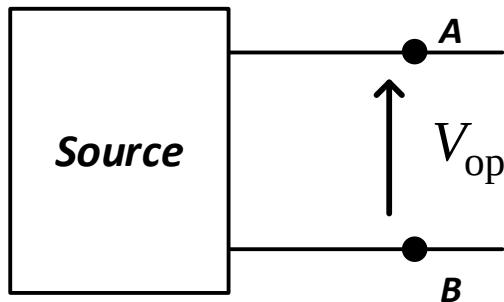
Thevenin Equivalent of the generalized Source



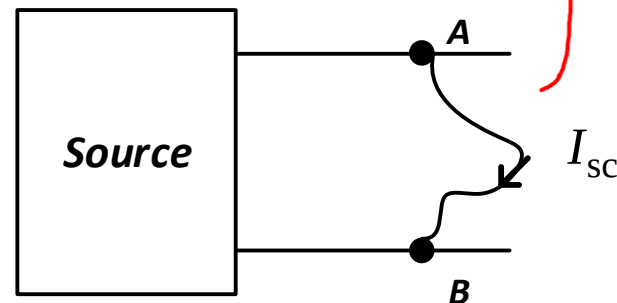
The condition for equivalence

$$\begin{cases} V_{th} = V_{op} \\ \frac{V_{th}}{Z_{th}} = I_{sc} \end{cases}$$

Applicable to complex number

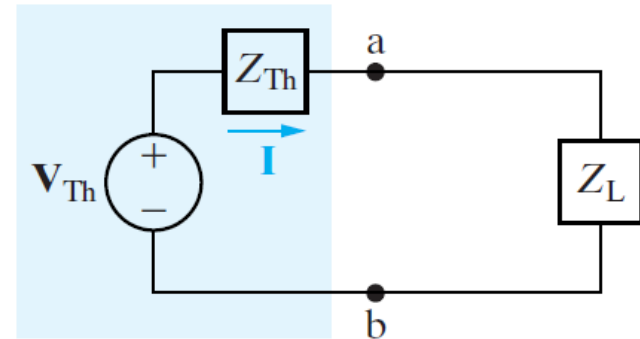
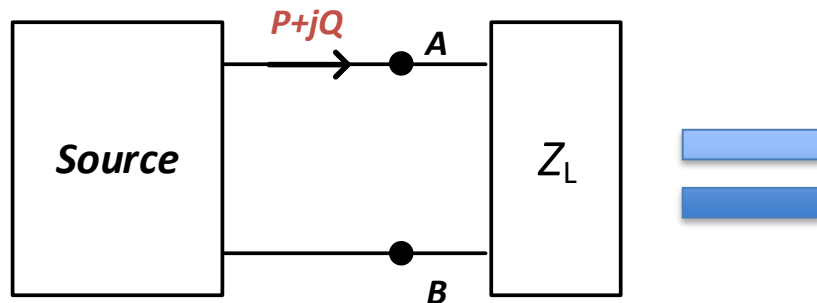


Open loop voltage



Short circuit current

Thevenin Equivalent of the generalized Source



Question: What is the load impedance Z_L that results in the delivery of maximum average power to terminals a and b.

Thevenin equivalent impedance

$$Z_{Th} = R_{Th} + jX_{Th}$$

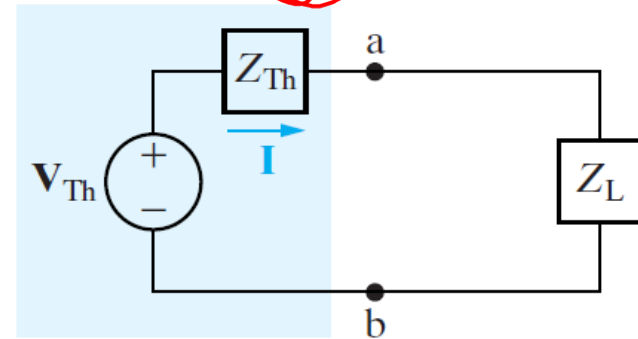
Load impedance

$$Z_L = R_R + jX_L$$

Active power consumed by the load:

$$P_L = |\mathbf{I}|^2 R_L$$

RMS value



The problem is reduced to find out the current in relation to the source and load impedance.

What $\mathbf{Z}_L$ results in $P_L^{\max}$?

$$\mathbf{I} = \frac{\mathbf{V}_{Th}}{\mathbf{Z}_{Th} + \mathbf{Z}_L} = \frac{\mathbf{V}_{Th}}{(\underline{R_{Th} + R_L}) + j(\underline{X_{Th} + X_L})}$$

$$|\mathbf{I}|^2 = \frac{|\mathbf{V}_{Th}|^2}{|(R_{Th} + R_L) + j(X_{Th} + X_L)|^2} = \frac{|\mathbf{V}_{Th}|^2}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

Maximum Power Transfer

We need to prove this!

$$Z_L = Z_{Th}^*.$$

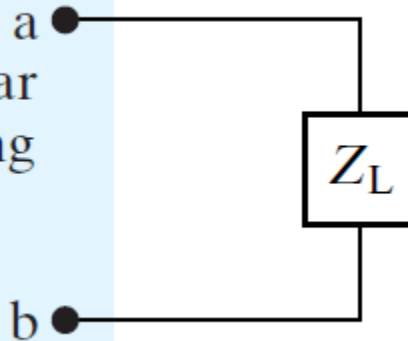
$$Z_{Th} = R_{Th} + jX_{Th},$$

$$Z_L = R_L + jX_L.$$

$$\mathbf{I} = \frac{\mathbf{V}_{Th}}{(R_{Th} + R_L) + j(X_{Th} + X_L)}.$$

$$P = |\mathbf{I}|^2 R_L.$$

Generalized linear network operating in the sinusoidal steady state



We replace this by its Thevenin equivalent

Determine the load impedance Z_L that results in the delivery of maximum average power to terminals a and b.

$$|I|^2 = \frac{|V_{Th}|^2}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

$$P_L = |I|^2 R_L$$



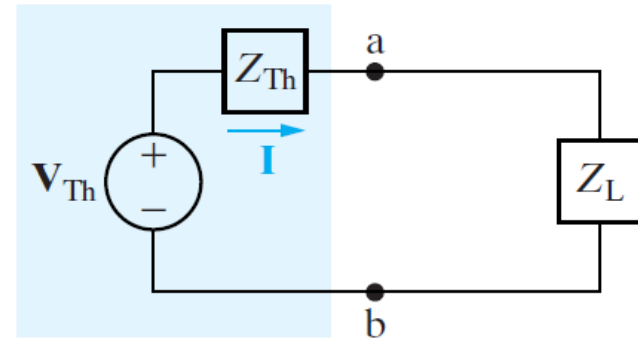
$$P_L = \frac{|V_{Th}|^2 R_L}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

V_{Th}, Z_{Th} are given
 Z_L is variable



$$P_L(R_L, X_L)$$

Load active power
 is a function of
 R_L and X_L



What Z_L results in $P_L^{\max}$?

Find the extreme value of a continuous function

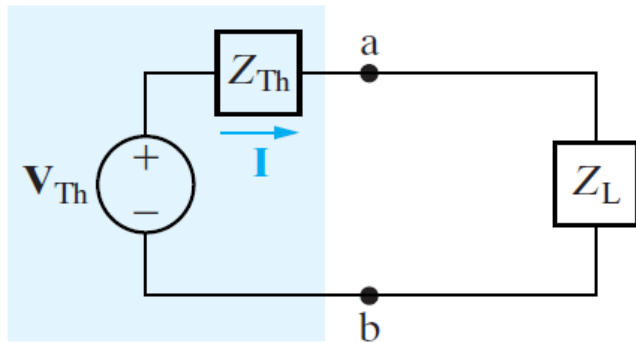
How to find the max value of $P_L(R_L, X_L)$?

$$f(x) = \frac{g(x)}{h(x)}$$

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{[h(x)]^2} = 0$$



x at which $f'(x) = 0$ will result in $f_{\max}(x)$



$$P = \frac{|\mathbf{V}_{Th}|^2 R_L}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}.$$

$g(x)$

$h(x)$

$$\frac{\partial P}{\partial R_L} = \frac{|\mathbf{V}_{Th}|^2 [(R_L + R_{Th})^2 + (X_L + X_{Th})^2 - 2R_L(R_L + R_{Th})]}{[(R_L + R_{Th})^2 + (X_L + X_{Th})^2]^2}.$$

$$\frac{\partial P}{\partial X_L} = \frac{-|\mathbf{V}_{Th}|^2 2R_L(X_L + X_{Th})}{[(R_L + R_{Th})^2 + (X_L + X_{Th})^2]^2},$$

$g'(x) = 0$
 $g(x) \cdot h'(x)$

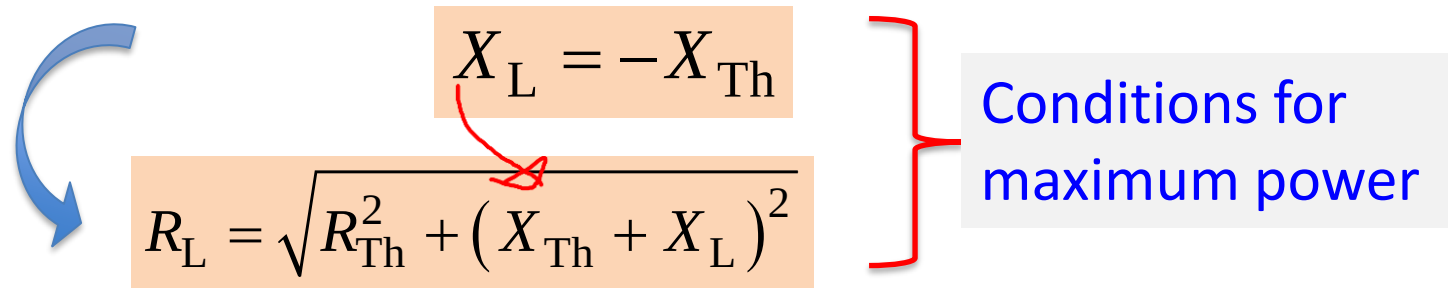
$= 0$
 $R_L \neq 0$

$$X_L = -X_{Th}.$$

$$R_L = \sqrt{R_{Th}^2 + (X_L + X_{Th})^2}.$$

Conditions for
maximum power

The condition for $P_{\max}$ is


$$X_L = -X_{Th}$$
$$R_L = \sqrt{R_{Th}^2 + (X_{Th} + X_L)^2}$$

Conditions for maximum power


$$Z_L = R_{Th} - jX_{Th} = (Z_{Th})^*$$

The maximum power delivered to the load occurs at $Z_L = (Z_{Th})^*$

Maximum Average Power Absorbed

$$P = |\mathbf{I}|^2 R_L.$$

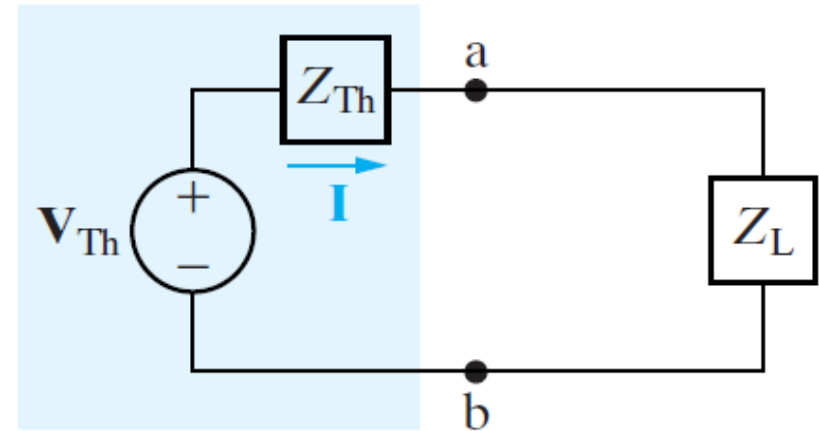
The *rms* load current is:

$$I_{rms} = \frac{V_{Th}}{2R_L}$$

$$P_{max} = \frac{|\mathbf{V}_{Th}|^2 R_L}{4R_L^2} = \frac{1}{4} \frac{|\mathbf{V}_{Th}|^2}{R_L}.$$

$$P_{max} = \frac{1}{8} \frac{V_m^2}{R_L}.$$

Peak value of voltage



Maximum power condition

$$Z_L = (Z_{Th})^*$$

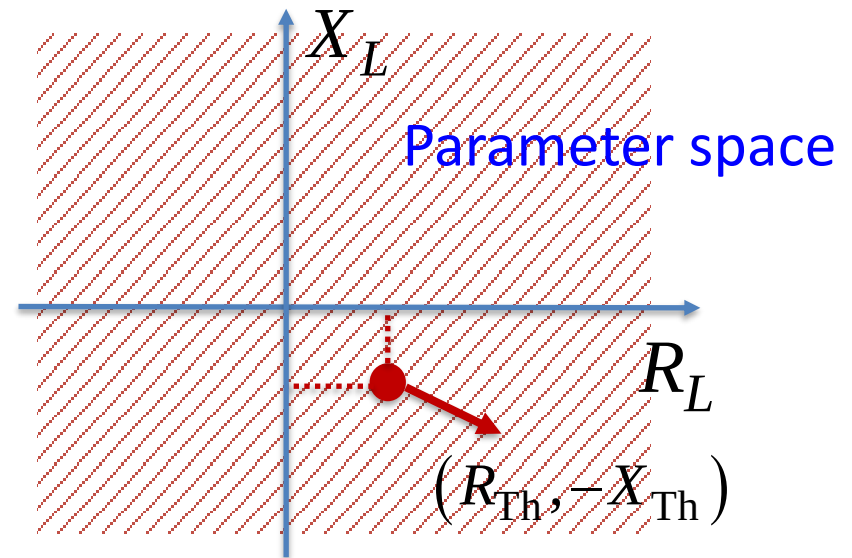
$$\begin{aligned} & \frac{Z_L + Z_{Th}}{=} \\ &= R_{Th} - \cancel{jX_{Th}} + R_{Th} + \cancel{jX_{Th}} \\ &= 2R_{Th} = 2R_L \end{aligned}$$

Summary of Maximum Power Transfer under Non-restrictive load impedence

Maximum power condition:

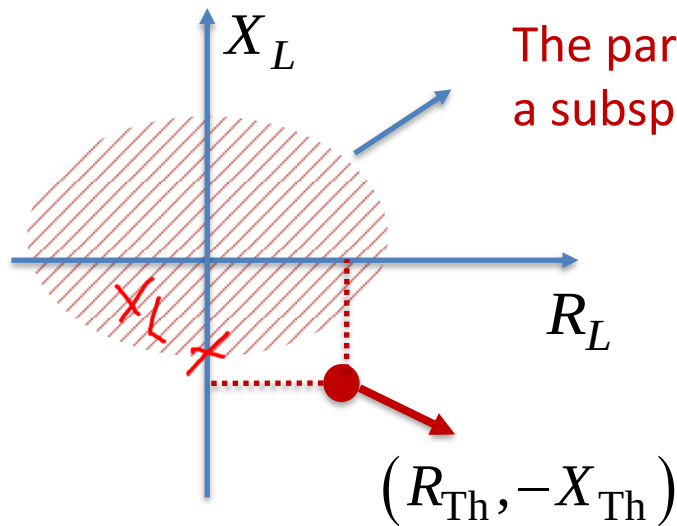
$$X_L = -X_{Th}.$$

$$R_L = \sqrt{R_{Th}^2 + (X_L + X_{Th})^2}.$$



no restrictions on R_L and X_L

What if we have a **restrictive** load impedance Z_L ?



The parameter space becomes a subspace of the whole plane

Conditions for non-restrictive load:

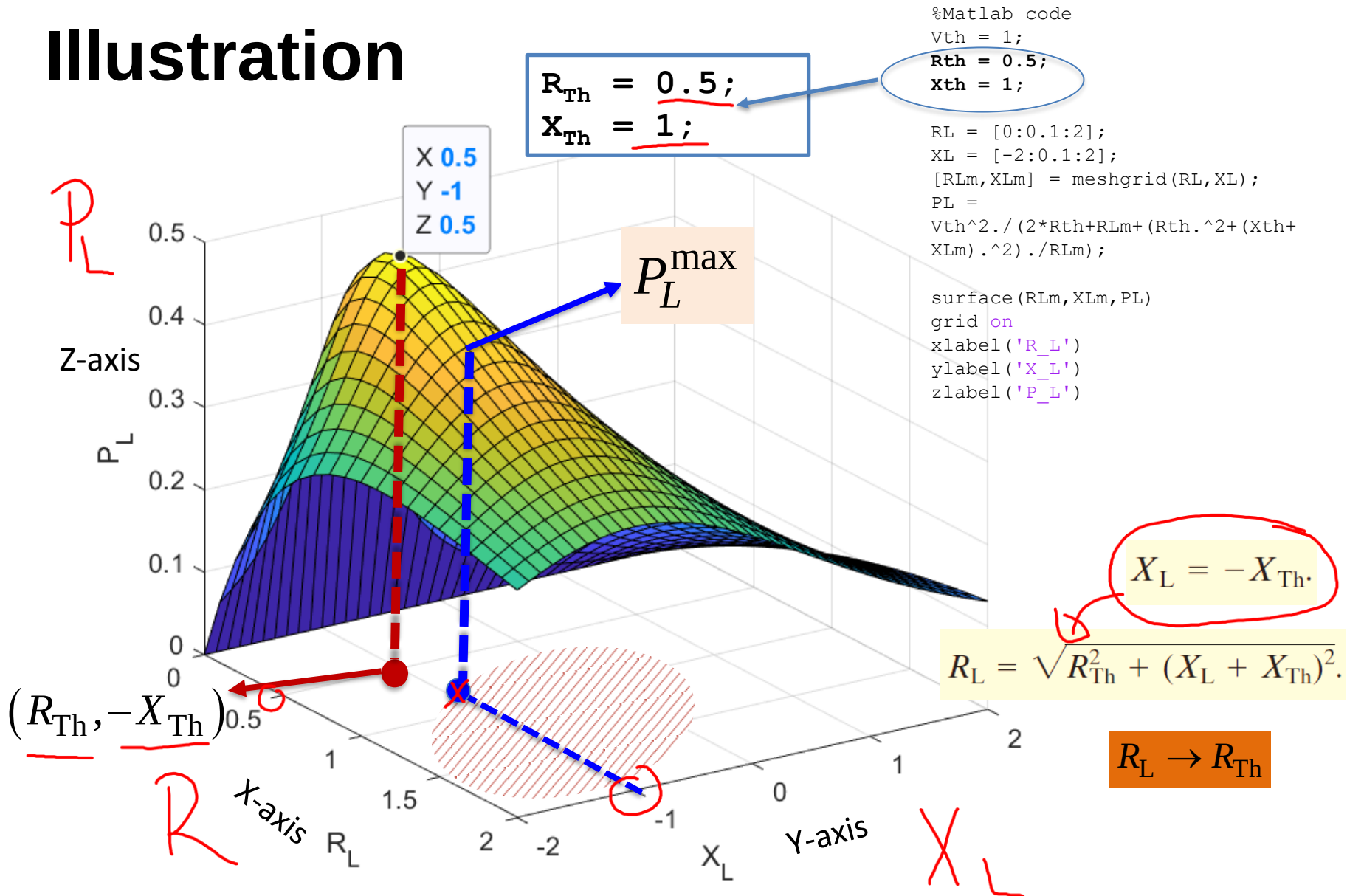
Condition 1: $X_L = -X_{Th}$

Condition 2: $R_L = \sqrt{R_{Th}^2 + (X_L + X_{Th})^2}$

The optimum condition for R_L and X_L is to adjust:

- X_L as near to $-X_{Th}$ as possible (condition 1)
- then adjust R_L as close to the condition 2 as possible .

Illustration



Restrictive Load

A Typical restrictive load Z_L

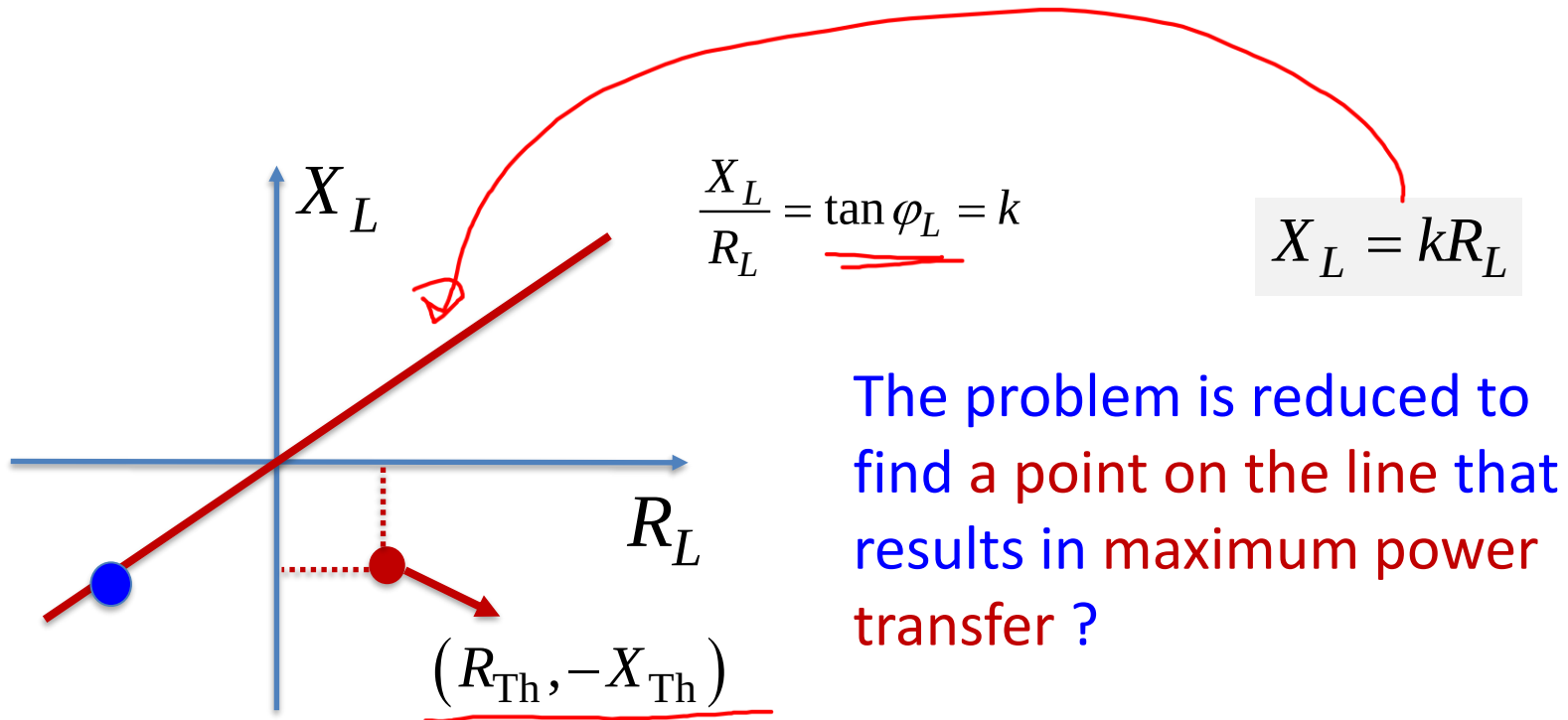
The **magnitude of Z_L can be varied** but its **phase angle cannot**,
what is the **condition** for the maximum power transfer ?

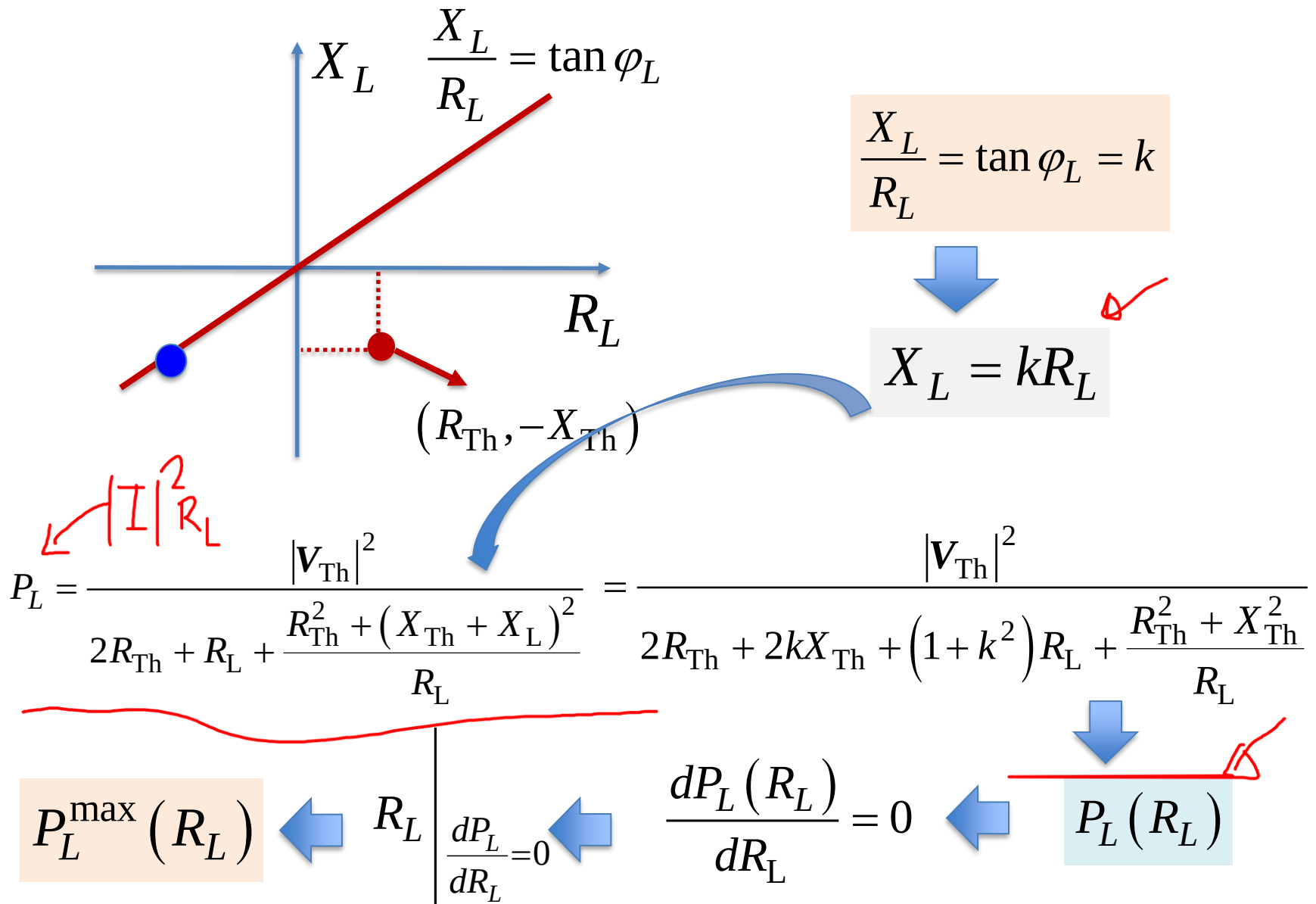
$$Z_L = \underbrace{|Z_L|}_{\text{Variable}} e^{j\underbrace{\varphi_L}_{\text{Fixed value}}}$$

$$Z_L = \underbrace{|Z_L| \cos \varphi_L}_{R_L} + j \underbrace{|Z_L| \sin \varphi_L}_{X_L} \quad \Rightarrow \quad \frac{X_L}{R_L} = \tan \varphi_L = k$$

Since X_L and R_L are correlated, we only have one variable needed to be determined.

What is the parameter space of this restrictive load ?





Calculating the $P_{\max}$

$$P_L = \frac{|V_{Th}|^2}{2R_{Th} + 2kX_{Th} + (1+k^2)R_L + \frac{R_{Th}^2 + X_{Th}^2}{R_L}}$$

$$P_L = \frac{|V_{Th}|^2}{2R_{Th} + 2kX_{Th} + (1+k^2) \left(R_L + \frac{(R_{Th}^2 + X_{Th}^2)/(1+k^2)}{R_L} \right)}$$

$$\frac{dP_L(R_L)}{dR_L} = 0$$

↓ ↓

$$X_L = kR_L$$

$$P_L = \frac{|V_{Th}|^2}{2R_{Th} + 2kX_{Th} + (1+k^2) \left(R_L + \frac{(R_{Th}^2 + X_{Th}^2)/(1+k^2)}{R_L} \right)}$$

The condition that enables the maximum value of P_L is

$$\underbrace{R_L = \frac{(R_{Th}^2 + X_{Th}^2)/(1+k^2)}{R_L}} \quad \Rightarrow \quad \underbrace{R_L^2 = \frac{R_{Th}^2 + X_{Th}^2}{(1+k^2)}}$$

$$\underbrace{X_L = kR_L} \quad \Rightarrow \quad \underbrace{X_L^2 = k^2 R_L^2 = k^2 \frac{R_{Th}^2 + X_{Th}^2}{(1+k^2)}}$$

Maximum power condition for this restrictive load

$$\left\{ \begin{array}{l} R_L^2 = \frac{R_{Th}^2 + X_{Th}^2}{(1+k^2)} \\ X_L^2 = k^2 R_L^2 = k^2 \frac{R_{Th}^2 + X_{Th}^2}{(1+k^2)} \end{array} \right.$$

$|Z_{Th}|^2$

$$\left\{ \begin{array}{l} R_L^2 = \frac{|Z_{Th}|^2}{(1+k^2)} \\ X_L^2 = k^2 R_L^2 = k^2 \frac{|Z_{Th}|^2}{(1+k^2)} \end{array} \right.$$

$$|Z_L|^2 = R_L^2 + X_L^2 = \frac{(1+k^2)|Z_{Th}|^2}{(1+k^2)} = |Z_{Th}|^2$$

$$|Z_L| = |Z_{Th}|$$

Summary of the **Conditions** for the **Maximum Power Transfer** to the Load

Non-restrictive load

$$Z_L = Z_{Th}^*$$

Phase-restrictive load

$$|Z_L| = |Z_{Th}|$$

When Z is restricted

a) When the load impedance is restricted to a set of resistance and reactance values, the optimum condition for R_L and X_L is to adjust X_L as near to as possible to $-X_{Th}$ and then adjust R_L as close to as possible to:

$$R_L = \sqrt{R_{Th}^2 + (X_L + X_{Th})^2}.$$

b) When the magnitude of Z_L can be varied but its phase angle cannot. Under this restriction the magnitude of Z_L is set equal to the magnitude of Z_{Th}

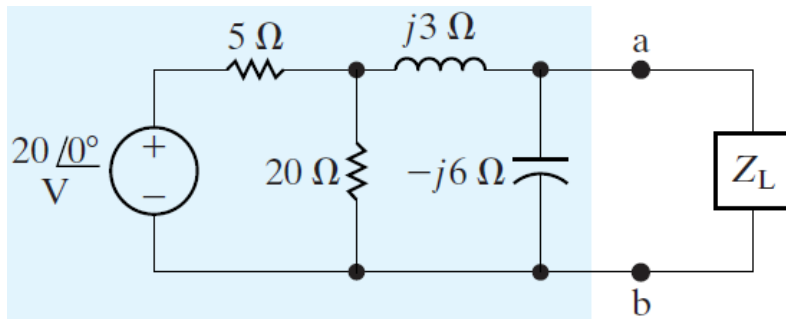
$$|Z_L| = |Z_{Th}|.$$

Practical Example

Maximum power transfer without Z_L restriction

For the circuit below determine:

- a) The impedance that results in maximum average power transfer to Z_L
- b) What is the maximum average power transferred to the load impedance determined in a?



$$Z_L = Z_{Th}^*$$

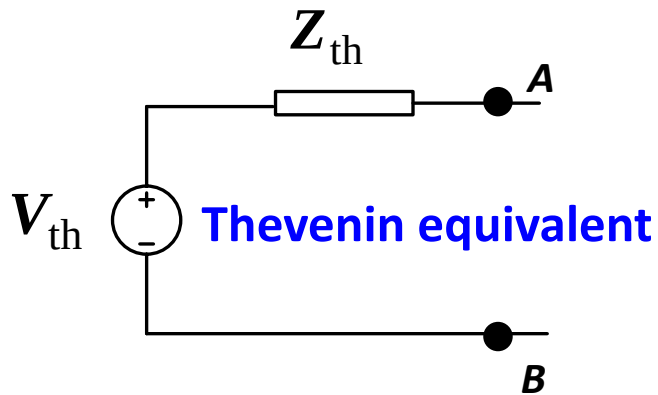
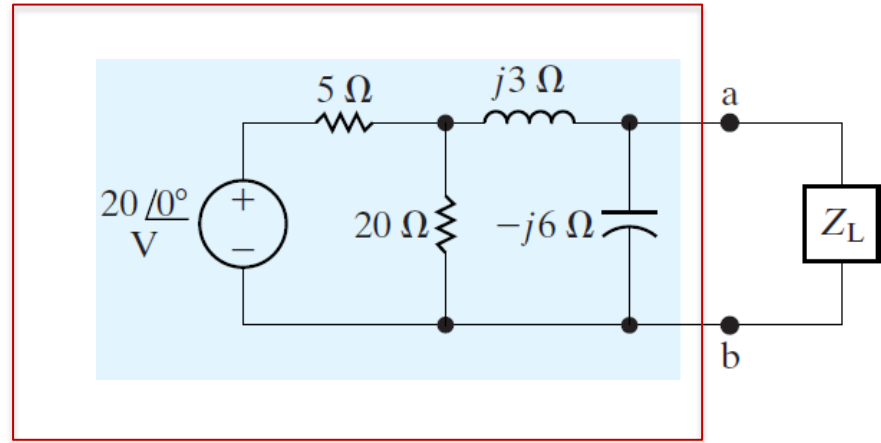
$$P = |I|^2 R_L$$

$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2 R_L}{4R_L^2} = \frac{1}{4} \frac{|\mathbf{V}_{Th}|^2}{R_L}$$

Answer: $Z_L = 5.76 + j1.68 \Omega$.

$$P = I_{\text{eff}}^2(5.76) = 8 \text{ W}.$$

What is the Thevenin equivalent of the source system ?

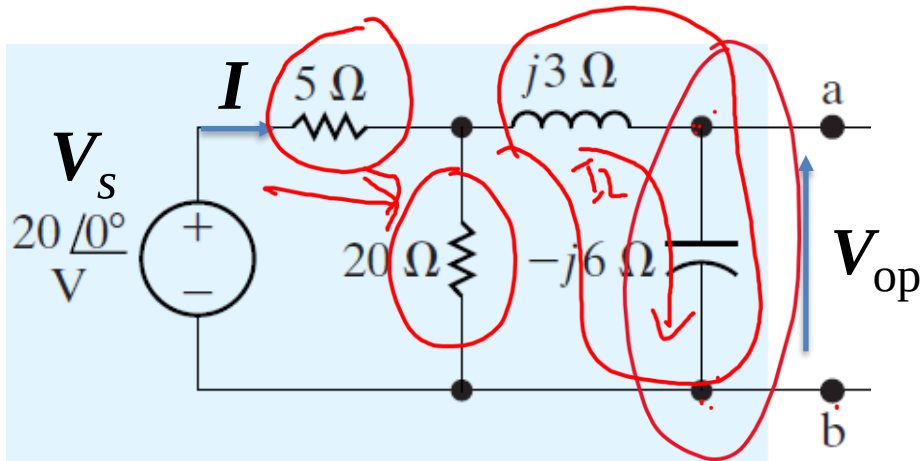


That has the same terminal characteristic as the original system.

Terminal characteristic =

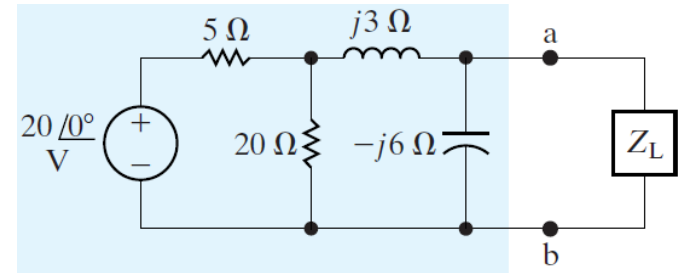
short circuit current + open loop voltage

Open Loop Voltage



$$Z_{eq} = 5\Omega + 20\Omega \parallel (3j - 6j)\Omega$$

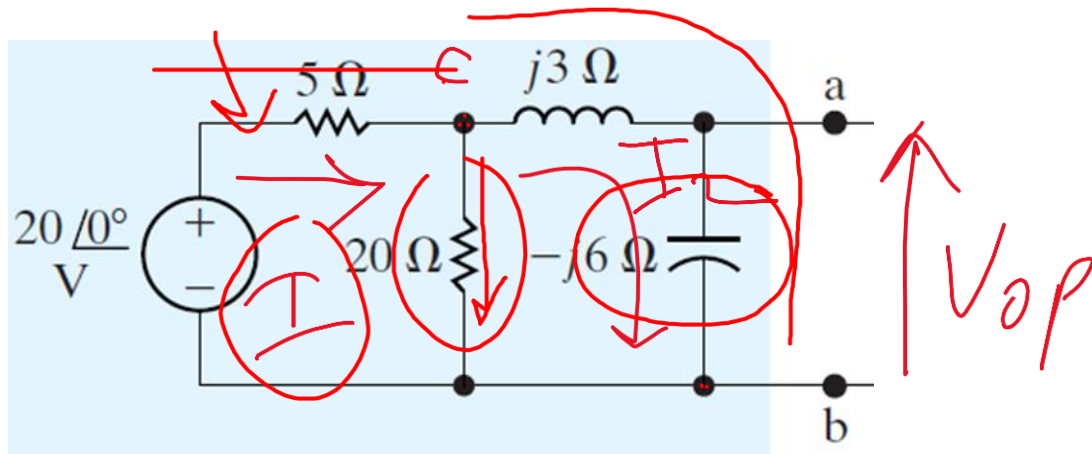
$$= 5 + \frac{20(-3j)}{20 - 3j} = 5.44 - 2.94j$$



$$I = \frac{V_s}{Z_{eq}} = \frac{20}{5.44 - 2.94j} = 3.24 \angle 28.4^\circ$$

$$V_{op} = \frac{I \cdot 20\Omega \parallel (3j - 6j)\Omega}{(3j - 6j)\Omega} \cdot (-6j)$$

$$= -19.2 \angle 126.93^\circ$$



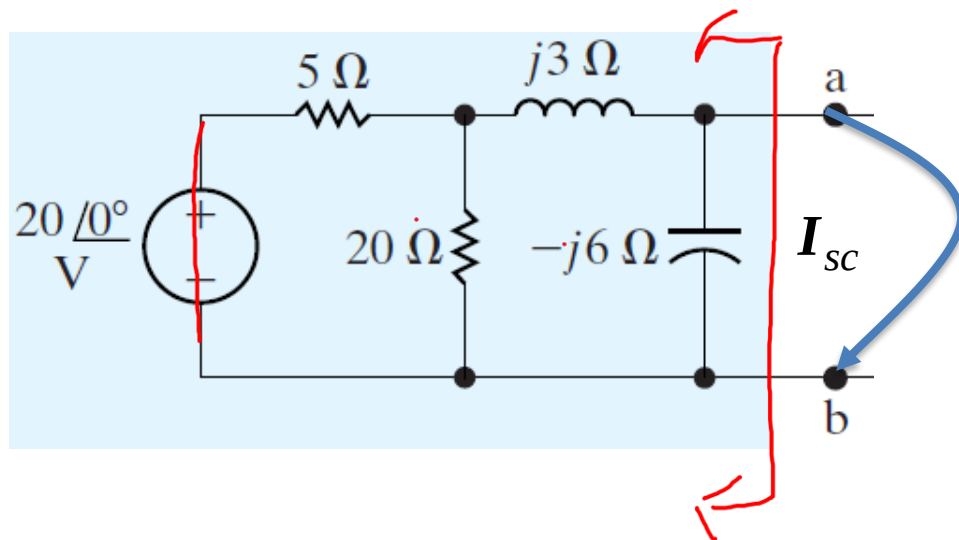
$$\underline{V_{cb}} = I_2 (-j3)$$

$$\underline{I_2 =}$$

$$V_{cb} = I \times \frac{(20 \times (-j3))}{20 - j3}$$

$$V_{op} = I_2 (-j6)$$

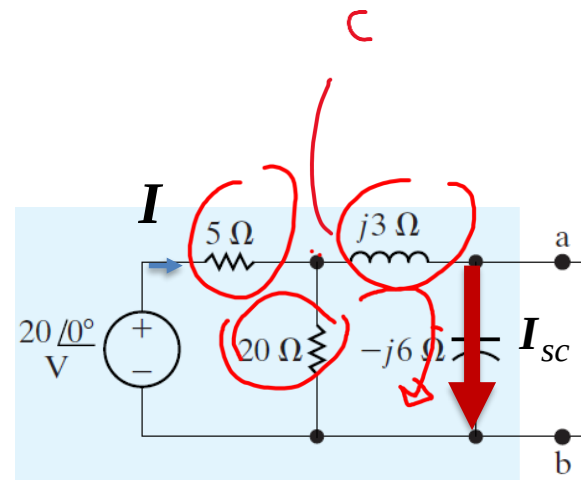
Short Circuit Current



$$Z_{eq} = 5\Omega + j3\Omega \parallel 20\Omega$$

$$= 5\Omega + \frac{j3\Omega \times 20\Omega}{j3\Omega + 20\Omega} = \underline{5.44 + 2.94j}$$

$$I = \frac{V_s}{Z_{eq}} = \frac{20}{5.44 + 2.94j} = \underline{3.236 \angle -28.4^\circ}$$



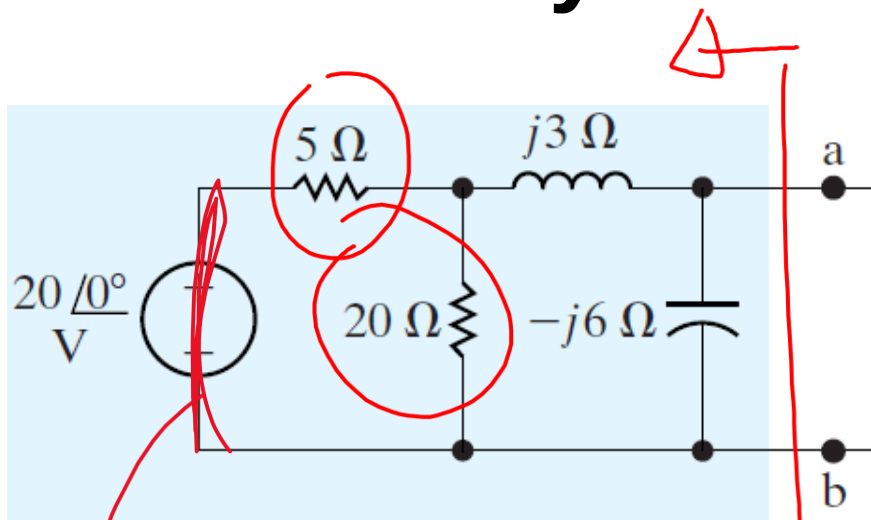
$$V_{cb} = I \times Z_{cb}$$

$$V_{cb} = I_{sc} \times 3j$$

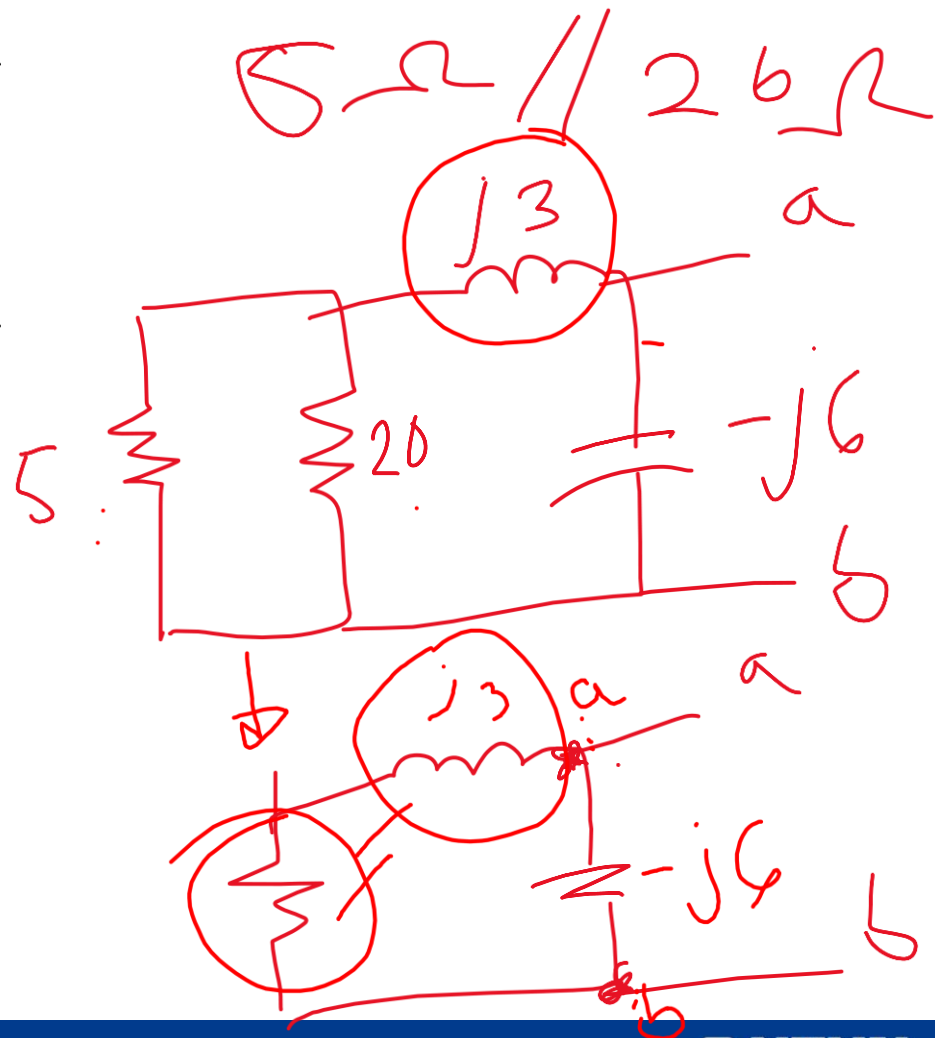
$$I_{sc} = \frac{I(20\Omega \parallel 3j\Omega)}{3j\Omega}$$

$$= I \frac{20}{20 + 3j} = \underline{3.2 \angle -36.93^\circ}$$

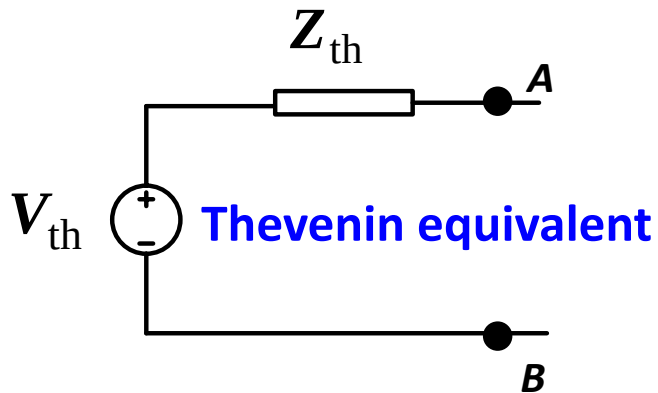
Another way



short circuit



Voltage and Impedance of the Thevenin equivalent Source



$$V_{Th} = V_{op} = -19.2 \angle 126.93^\circ$$

$$\frac{V_{Th}}{Z_{Th}} = I_{sc}$$

$$Z_{Th} = \frac{V_{Th}}{I_{sc}} = \frac{-19.2 \angle 126.93^\circ}{3.2 \angle -36.93^\circ} = -6 \angle 163.86^\circ$$

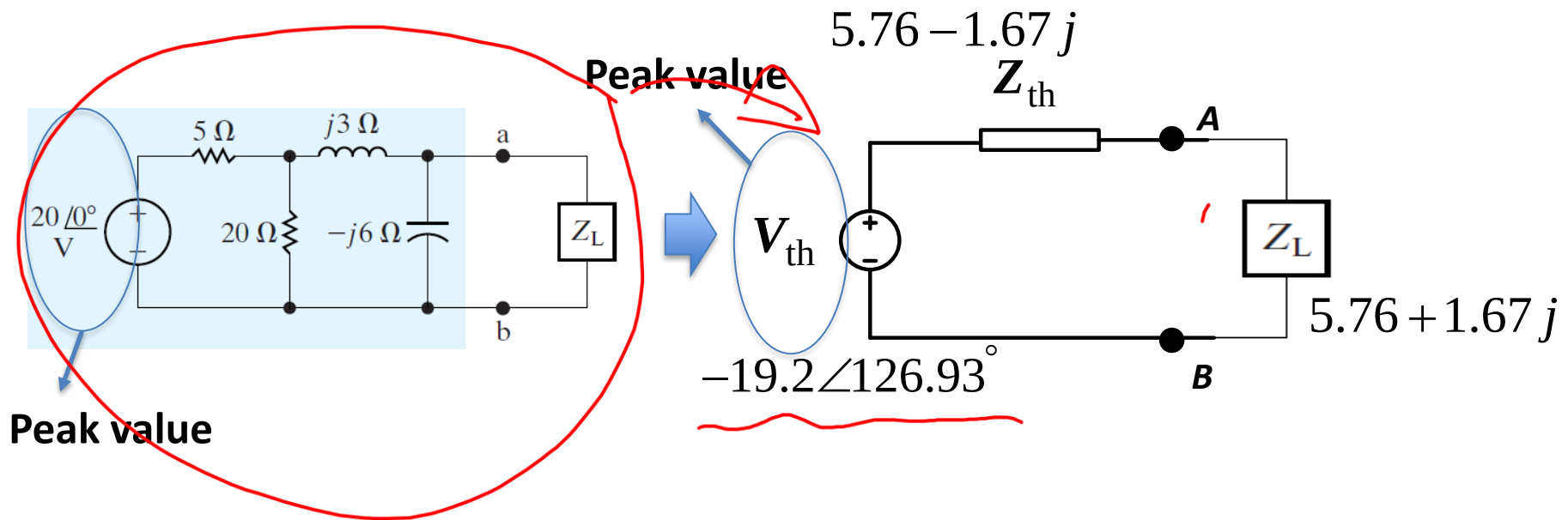
$$= (5.76 - 1.67j) \Omega$$

- a) What value of Z_L results in maximum average power transfer and what is the maximum power?

Non-restrictive condition:

$$Z_L = Z_{Th}^* = (5.76 + 1.67j) \Omega$$

Thevenin equivalent of the original system

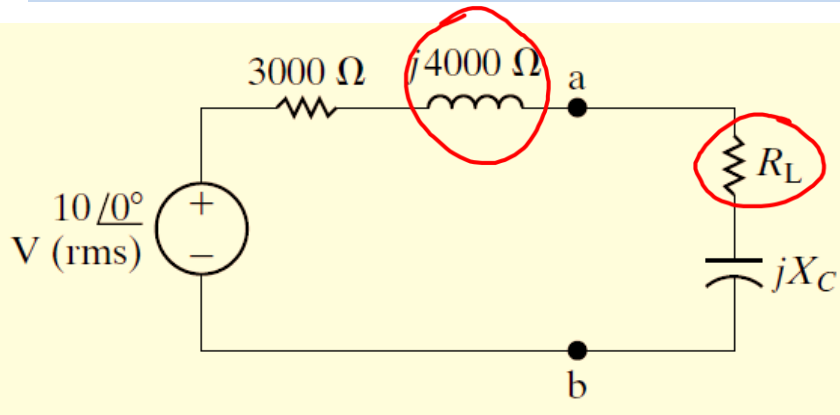


$$P_L^{\max} = \frac{|V_{Th}|^2}{8R_L} = \frac{19.2^2}{8 \times 5.76} = 8W$$

Maximum power transfer with Z_L restriction

For the circuit below determine:

- What value of Z_L results in maximum average power transfer and what is the maximum power?
- If R_L can be varied between 0 and 4000 Ω and X_C between 0 and -2000 Ω , what values of R_L and X_L transfer the most average power to the load? And what is the maximum average power under such restrictions?



$$Z_L = Z_{Th}^*$$

$$Z_L = 3000 - j4000 \Omega.$$

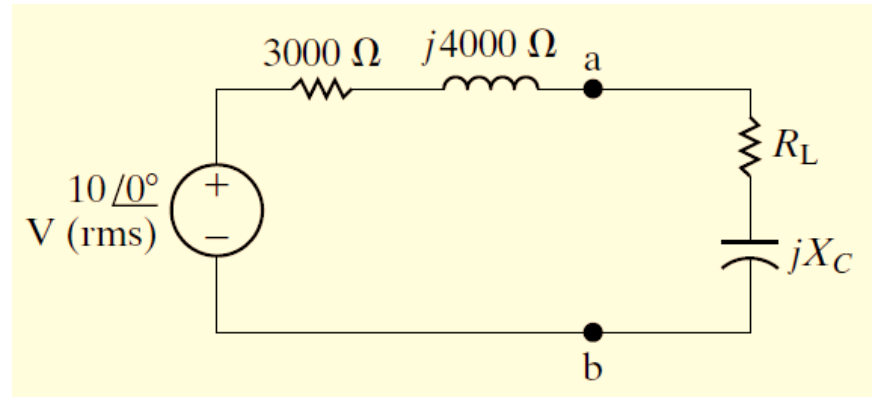
$$P = \frac{1}{4} \frac{10^2}{3000} = \frac{25}{3} \text{ mW} = 8.33 \text{ mW}.$$

Answer:

$$Z_L = 3605.55 - j2000 \Omega.$$

$$P = (1.4489 \times 10^{-3})^2 (3605.55) = 7.57 \text{ mW}.$$

a) What value of Z_L results in maximum average power transfer and what is the maximum power?



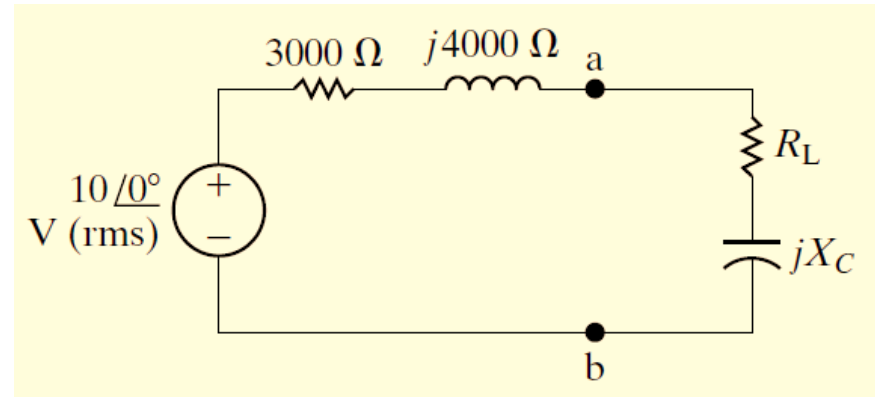
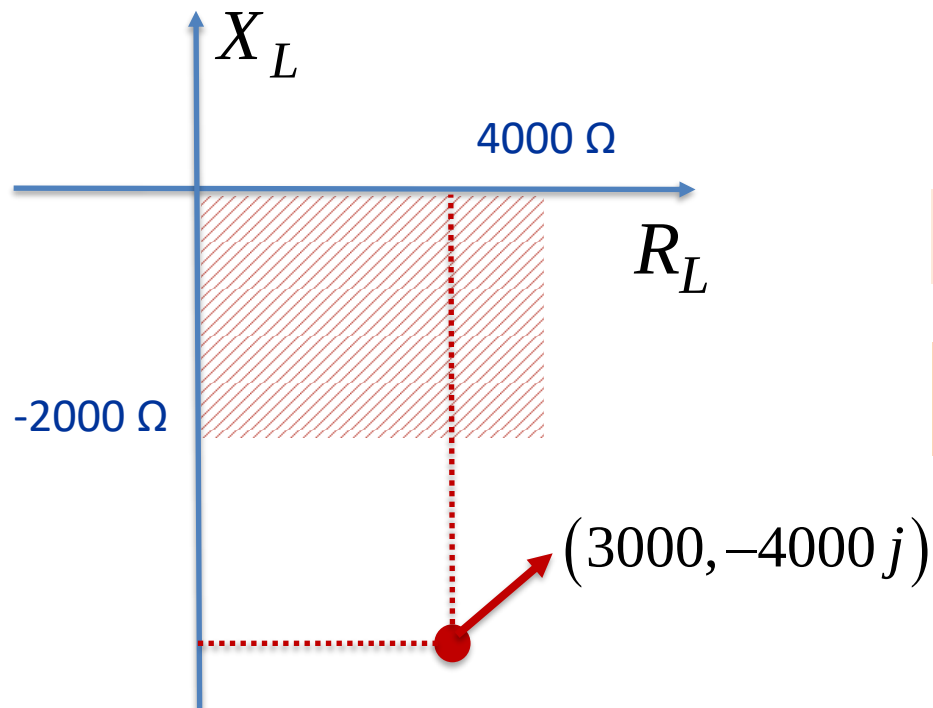
$$\mathbf{Z}_{Th} = 3000\Omega + 4000j\Omega$$

$$\mathbf{V}_{Th} = 10\angle 0^\circ, RMS$$

$$\mathbf{Z}_L = \mathbf{Z}_{Th}^* = (3000 - 4000j)\Omega$$

$$P_{Lmax} = \frac{|\mathbf{V}_{Th}|^2}{4R_L} = \frac{10^2}{4 \times 3000} = 8.33mW$$

b) If R_L can be varied between 0 and $4000\ \Omega$ and X_C between 0 and $-2000\ \Omega$, what values of R_L and X_L transfer the most average power to the load? And what is the maximum average power under such restrictions?



condition of **unrestrictive** load

$$\mathbf{Z}_L = \mathbf{Z}_{Th}^* = (3000 - 4000j)\ \Omega$$

$$X_L \rightarrow -X_{Th}$$



$$X_C = -2000$$

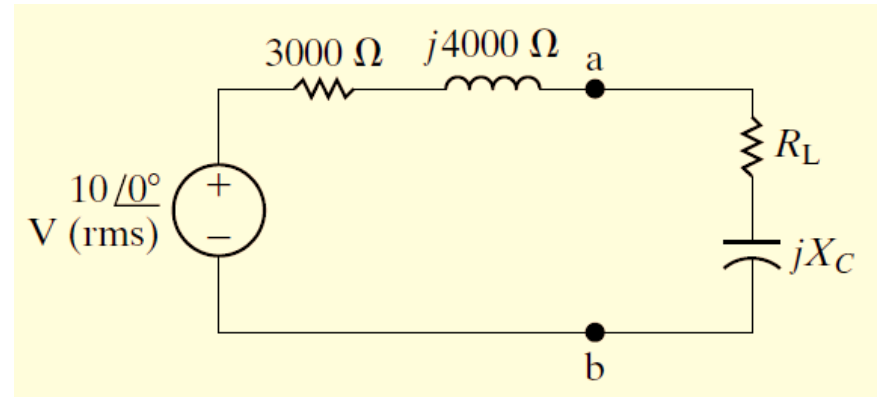
$$R_L \rightarrow \sqrt{R_{Th}^2 + (X_{Th} + X_C)^2}$$

$$= \sqrt{3000^2 + 2000^2} = 3605.5$$



$$R_L = 3605.5$$

b) what is the maximum average power under such restrictions?



$$X_C = -2000$$

$$R_L = 3605.5$$

$$\mathbf{Z}_L = (3605.5 - 2000j) \Omega$$

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}_L + \mathbf{Z}_{Th}} = \frac{10(rms)}{6605.5 + 2000j} = 1.45 \angle -16.85^\circ \text{ mA}$$

$$P_L^{\max} = |\mathbf{I}|^2 R_L = (1.45 \times 10^{-3})^2 \times 3605.5 = 7.58 \text{ mW}$$

End